

Multimodal Datasets for the Bemba Language Datasheet

I. MOTIVATION FOR DATASHEET CREATION

A. Why was the datasheet created? (e.g., was there a specific task in mind? was there a specific gap that needed to be filled?)

The dataset was created to enable the development of speech recognition and text/speech to text translation applications for the Bemba language, as well as facilitate research in language grounding and multimodal model development.

B. Who created the dataset (e.g., which team, research group) and on behalf of which entity (e.g., company, institution, organization)?

The dataset was created by Claytone Sikasote (University of Zambia), Eunice M. Mulenga (University of Zambia) and Antonios Anastasopoulos (George Mason University) at University of Zambia.

C. Has the dataset been used already? If so, where are the results so others can compare (e.g., links to published papers)?

In progress

D. What (other) tasks could the dataset be used for?

We envision that the dataset could also be used image grounded conversations task.

E. Who funded the creation dataset?

The project to create the dataset was funded by the LacunaFund under the project grant number: 2020-20-055(19397.70).

F. Any other comment?

N/A

II. DATASHEET COMPOSITION

A. What are the instances?(that is, examples; e.g., documents, images, people, countries) Are there multiple types of instances? (e.g., movies, users, ratings; people, interactions between them; nodes, edges)

The instances are image grounded conversation Bemba transcriptions with corresponding English translations.

B. How many instances are there in total (of each type, if appropriate)?

There are 92, 182 instances in terms of number of text sentences/audio utterances amounting to 187 hours of speech data from a total of 16, 214 images.

C. What data does each instance consist of ? “Raw” data (e.g., unprocessed text or images)? Features/attributes? Is there a label/target associated with instances? If the instances related to people, are subpopulations identified (e.g., by age, gender, etc.) and what is their distribution?

Each instance has the audio utterance with its associated image, Bemba transcription and English translation.

D. Is there a label or target associated with each instance? If so, please provide a description.

Owing to the multiple-purpose and multimodality nature of the dataset, the labels are dependant on the task: for ASR, the label is the Bemba transcription; for MT and ST, the English translation are labels.

E. Is any information missing from individual instances? If so, please provide a description, explaining why this information is missing (e.g., because it was unavailable). This does not include intentionally removed information, but might include, e.g., redacted text.

All necessary information as extracted from the metadata as supplied by participants is included.

F. Are relationships between individual instances made explicit (e.g., users’ movie ratings, social network links)? If so, please describe how these relationships are made explicit.

Not explicitly, though the image, audio and all text information is organized in such a way that unique conversations can easily be extracted.

G. Does the dataset contain all possible instances or is it a sample (not necessarily random) of instances from a larger set? If the dataset is a sample, then what is the larger set? Is the sample representative of the larger set (e.g., geographic coverage)? If so, please describe how this representativeness was validated/verified. If it is not representative of the larger set, please describe why not (e.g., to cover a more diverse range of instances, because instances were withheld or unavailable).

The dataset is not a sample of any larger dataset. It contains all possible instances.

H. Are there recommended data splits (e.g., training, development/validation, testing)? If so, please provide a description of these splits, explaining the rationale behind them.

The dataset is split into train, validation and testing sets following the original Flickr30K[CITE] dataset splits.

I. Are there any errors, sources of noise, or redundancies in the dataset? If so, please provide a description.

There are X number of images that are duplicates.

J. Is the dataset self-contained, or does it link to or otherwise rely on external resources (e.g., websites, tweets, other datasets)? If it links to or relies on external resources, a) are there guarantees that they will exist, and remain constant, over time; b) are there official archival versions of the complete dataset (i.e., including the external resources as they existed at the time the dataset was created); c) are there any restrictions (e.g., licenses, fees) associated with any of the external resources that might apply to a future user? Please provide descriptions of all external resources and any restrictions associated with them, as well as links or other access points, as appropriate.

The dataset is self-contained.

Any other comments?

N/A

III. COLLECTION PROCESS

A. What mechanisms or procedures were used to collect the data (e.g., hardware apparatus or sensor, manual human curation, software program, software API)? How were these mechanisms or procedures validated?

The data was collected from participants who were recruited as speakers to record multi-turn conversations using The LIG-AIKUMA[CITE] mobile applications.

B. How was the data associated with each instance acquired? Was the data directly observable (e.g., raw text, movie ratings), reported by subjects (e.g., survey responses), or indirectly inferred/derived from other data (e.g., part-of-speech tags, model-based guesses for age or language)? If data was reported by subjects or indirectly inferred/derived from other data, was the data validated/verified? If so, please describe how.

The data was collected from participants who were recruited as speakers to record multi-turn conversations using The LIG-AIKUMA mobile applications.

C. If the dataset is a sample from a larger set, what was the sampling strategy (e.g., deterministic, probabilistic with specific sampling probabilities)?

The dataset is not a sample of a larger dataset.

D. Who was involved in the data collection process (e.g., students, crowdworkers, contractors) and how were they compensated (e.g., how much were crowdworkers paid)?

We crowd-sourced workers for our data collection of process; speakers, transcribers, translators and validators. Each participant was paid \$20 dollars/hour spent on the task for a specified amount of target work.

E. Over what timeframe was the data collected? Does this timeframe match the creation timeframe of the data associated with the instances (e.g., recent crawl of old news articles)? If not, please describe the timeframe in which the data associated with the instances was created.

The dataset was collected over the period of one year.

IV. DATA PREPROCESSING

A. Was any preprocessing/cleaning/labeling of the data done (e.g., discretization or bucketing, tokenization, part-of-speech tagging, SIFT feature extraction, removal of instances, processing of missing values)? If so, please provide a description. If not, you may skip the remainder of the questions in this section.

N/A

B. Was the “raw” data saved in addition to the preprocessed/cleaned/labeled data (e.g., to support unanticipated future uses)? If so, please provide a link or other access point to the “raw” data.

N/A

C. Is the software used to preprocess/clean/label the instances available? If so, please provide a link or other access point.

N/A

D. Does this dataset collection/processing procedure achieve the motivation for creating the dataset stated in the first section of this datasheet? If not, what are the limitations?

N/A

E. Any other comments

N/A

V. DATASET DISTRIBUTION

A. How will the dataset be distributed? (e.g., tarball on website, API, GitHub; does the data have a DOI and is it archived redundantly?)

The dataset is publicly available on the following project website link: **LINK TO BE ADDED**

B. When will the dataset be released/first distributed? What license (if any) is it distributed under?

The will be released on X.Y.Z under the Creative Commons BY-NC-ND 4.0 license.

C. Are there any copyrights on the data?

TBA

D. Are there any fees or access/export restrictions?

No fees or access restrictions!

E. Any other comments?

N/A

VI. DATASET MAINTENANCE

A. Who is supporting/hosting/maintaining the dataset?

Claytone Sikasote is supporting/maintaining the dataset.

B. Will the dataset be updated? If so, how often and by whom?

Yes the dataset will be updated occasionally.

C. How will updates be communicated? (e.g., mailing list, GitHub)

Updates will be communicated via Github.

D. If the dataset becomes obsolete how will this be communicated?

N/A

E. Is there a repository to link to any/all papers/systems that use this dataset?

TO BE UPDATED

F. If others want to extend/augment/build on this dataset, is there a mechanism for them to do so? If so, is there a process for tracking/assessing the quality of those contributions. What is the process for communicating/distributing these contributions to users?

via GitHub

VII. LEGAL AND ETHICAL CONSIDERATIONS

A. Were any ethical review processes conducted (e.g., by an institutional review board)? If so, please provide a description of these review processes, including the outcomes, as well as a link or other access point to any supporting documentation.

N/A

B. Does the dataset contain data that might be considered confidential (e.g., data that is protected by legal privilege or by doctor/patient confidentiality, data that includes the content of individuals non-public communications)? If so, please provide a description.

No, it does not.

C. Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety? If so, please describe why

No, it does not.

D. Does the dataset relate to people? If not, you may skip the remaining questions in this section.

N/A

E. Does the dataset identify any subpopulations (e.g., by age, gender)? If so, please describe how these subpopulations are identified and provide a description of their respective distributions within the dataset.

N/A

F. Is it possible to identify individuals (i.e., one or more natural persons), either directly or indirectly (i.e., in combination with other data) from the dataset? If so, please describe how.

No, the dataset does not contain any personal identifiable information.

G. Does the dataset contain data that might be considered sensitive in any way (e.g., data that reveals racial or ethnic origins, sexual orientations, religious beliefs, political opinions or union memberships, or locations; financial or health data; biometric or genetic data; forms of government identification, such as social security numbers; criminal history)? If so, please provide a description.

N/A

H. Did you collect the data from the individuals in question directly, or obtain it via third parties or other sources (e.g., websites)?

Directly

I. Were the individuals in question notified about the data collection? If so, please describe (or show with screenshots or other information) how notice was provided, and provide a link or other access point to, or otherwise reproduce, the exact language of the notification itself.

Yes, we recruited the participant that were involved in the data collection process directly.

J. Did the individuals in question consent to the collection and use of their data? If so, please describe (or show with screenshots or other information) how consent was requested and provided, and provide a link or other access point to, or otherwise reproduce, the exact language to which the individuals consented.

Yes, all participants signed consent form at the time of recruitment.

K. If consent was obtained, were the consenting individuals provided with a mechanism to revoke their consent in the future or for certain uses? If so, please provide a description, as well as a link or other access point to the mechanism (if appropriate).

N/A

L. Has an analysis of the potential impact of the dataset and its use on data subjects (e.g., a data protection impact analysis) been conducted? If so, please provide a description of this analysis, including the outcomes, as well as a link or other access point to any supporting documentation.

TO BE DETERMINED

M. Any other comments?

N/A

REFERENCES